The *way* that the ribs blur together is important in understanding the limitations of this measurement. Imagine acquiring an image of the line pair gauge in Fig. 25-2. Figures (a) and (b) show examples of the profiles at low and high spatial frequencies. At the low frequency, shown in (b), the curve is flat on the top and bottom, but the edges are blurred, At the higher spatial frequency, (a), the *amplitude* of the modulation has been reduced. This is exactly what the MTF curve in Fig. 25-1b describes: higher spatial frequencies are reduced in amplitude. The individual ribs will be distinguishable in the image as long as the amplitude is greater than about 3% to 10% of the original height. This is related to the eye's ability to distinguish the low contrast difference between the peaks and valleys in the presence of image noise.

A strong advantage of the line pair gauge measurement is that it is simple and fast. The strongest disadvantage is that it relies on the human eye, and therefore has a certain subjective component. Even if the entire MTF curve is measured, the most common way to express the system resolution is to quote the frequency where the MTF is reduced to either 3%, 5% or 10%. Unfortunately, you will not always be told which of these values is being used; product data sheets frequently use vague terms such as "limiting resolution." Since manufacturers like their specifications to be as good as possible (regardless of what the device actually does), be safe and interpret these ambiguous terms to mean 3% on the MTF curve.

A subtle point to notice is that the MTF is defined in terms of *sine* waves, while the line pair gauge uses *square* waves. That is, the ribs are uniformly dark regions separated by uniformly light regions. This is done for manufacturing convenience; it is very difficult to make lines that have a sinusoidally varying darkness. What are the consequences of using a square wave to measure the MTF? At high spatial frequencies, all frequency components but the fundamental of the square wave have been removed. This causes the modulation to appear sinusoidal, such as is shown in Fig. 25-2a. At low frequencies, such as shown in Fig. 25-2b, the wave appears square. The fundamental sine wave contained in a square wave has an amplitude of $4/\pi = 1.27$ times the amplitude of the square wave (see Table 13-10). The result: the line pair gauge provides a slight overestimate of the true resolution of the system, by starting with an effective amplitude of more than pure black to pure white. Interesting, but almost always ignored.

Since square waves and sine waves are used interchangeably to measure the MTF, a special terminology has arisen. Instead of the word "cycle," those in imaging use the term **line pair** (a dark line next to a light line). For example, a spatial frequency would be referred to as *25 line pairs per millimeter*, instead of *25 cycles per millimeter*.

The width of the PSF doesn't track well with human perception and is difficult to measure. The MTF methods are in the *wrong domain* for understanding how resolution affects the encoded information. Is there a more favorable alternative? The answer is yes, the **line spread function (LSF)** and the **edge response**. As shown in Fig. 25-3, the line spread